VPDA - Project assignment - 2026

Read carefully these instructions. There will be penalizations if your work does not meet the instructions.

Team Formation (3-4 students)

You are given the option to indicate ONE student with whom you want to work. (by 2/10/2026)

I will form the final teams (by 6/10/2026)

Task

The project requires you to:

1. Formulate a research question (or a set of related questions) and address it through **data visualization in R/ggplot2**. The question should be relevant, insightful, and clearly guide the overall direction of your analysis.
2. Design and create one or more non-trivial visualizations using ggplot2 in R. The visual representation should go beyond standard chart types and demonstrate deliberate design choices tailored to the research question and the characteristics of the data. Creativity should contribute to communicating the information effectively and should be explicitly justified in the report.
    - **Domain knowledge:** select a topic that at least one team member is familiar with or expect to spend time understanding the domain to define meaningful questions.
    - **Complexity**: the visual representations should capture multi-dimensional relationships, temporal trends, or comparisons across variables.
    - **Clarity**: make use of scales, colors, and annotations to enhance understanding.
    - **Creativity**: aim to integrate information innovatively, combining plots or using advanced techniques when appropriate.
    - **Informative**: create a visual representation that:
        1. clearly answers the research question(s) and provides meaningful insights.
        2. is visually appealing while serving the data's message.
        3. includes appropriate labeling, contrasts, and readability.
3. If you create multiple visualizations, you should combine them into a single cohesive image to highlight connections within the dataset.

Dataset

Choose a dataset that has the potential to highlight a significant issue in society, politics, economics, sports, or any other relevant domain. Example sources of data:

- The PORDATA website (https://www.pordata.pt/) provides numerous statistics about Portugal and Europe.
- The GPEARi website (https://www.gpeari.gov.pt/web/pt/indicadores-mensais-de-conjuntura) provides Monthly Economic Indicators about Portugal.
- The GitHub repository by Jeff Sackmann (https://github.com/JeffSackmann) offers a wide range of datasets on tennis and other sports.

You may also choose other datasets of interest or combine multiple datasets. Feel free to extend your dataset with external sources if relevant.

Visualization requirements

The final visualization must include (If multiple visualizations are created, they should be combined into a single image):

- A meaningful title.
- Authors' names in the caption or subtitle.
- Date of creation.

Report

The final report (in PDF format) should:

- Provide context for the topic being analyzed.
- Clearly state the question(s).
- Describe the dataset(s) used and any pre-processing steps involved.
- Justify the visualization design choices (e.g., chart types, encoding, colors, fonts), explaining why the chosen representation is appropriate, what makes it creative, and why alternative or more conventional representations were not considered as effective.
- Show the evolution of the visualization, including at least two substantially different intermediate versions, and explain the decisions that led from each version to the final design.
- Present and discuss the insights gained from the visualization(s), highlighting their significance and any potential limitations.
- Include at least one figure of the final visualization(s), in addition to the separate PNG file.
- A final page stating: (i) The contribution of team members. A short paragraph describing the specific contribution of each team member. (ii) Use of generative AI. A paragraph describing whether and how generative AI tools were used in the project, including the tasks for which they were used.

Page limit: 10 pages, using an 11-pt font (excluding references and appendices, if any).

Deliverables

Submit the following by email to jlborges@fe.up.pt:

(make sure that all your filenames start with your team ID)

1. The dataset file(s)
2. The R script file
3. The final report in PDF format (T0X_topic.pdf)
4. A high-resolution PNG image of the visualization(s) (suitable for printing)
5. CC all team members in the submission email
6. The email subject line must include: course name, team ID, and a reference to its contents (e.g., *VPDA – T03 – project deliverables*)

Final Remark

A very good project should reveal a clear process:

domain knowledge -> relevant question ->ma appropriate data -> visualization design -> insight

Generative AI may be used to support this process, but not to replace the knowledge, reasoning, and critical decisions expected from the students. If the submitted work raises doubts about the students' effective involvement or understanding of the project, additional verification procedures may be required and may affect the assessment.

Deadline: 30.10.2026 (Friday) at 23:59:59